WEEK 3 · RESP-100 BASIC EKG INTERPRETATION

Waves and Measurements

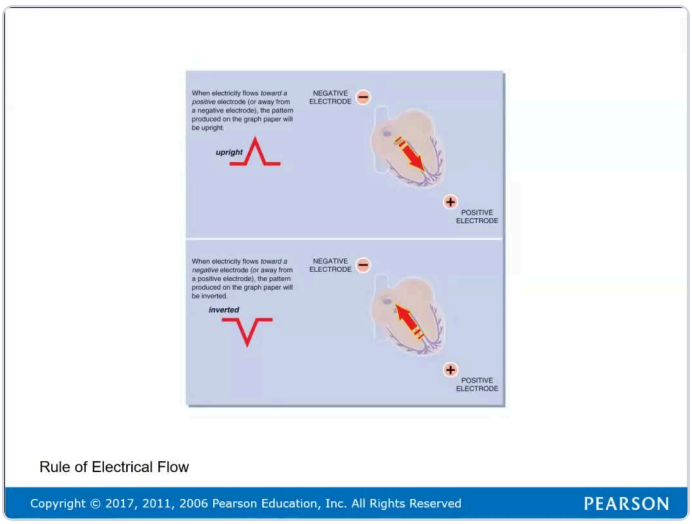
Walraven, *Basic Arrhythmias* 8th ed. · Chapter 2 (plus a review of Appendix C leads) · lecture 0:00–0:33

How to read EKG graph paper, what each wave means, and how to measure the PR interval and QRS by counting small boxes.

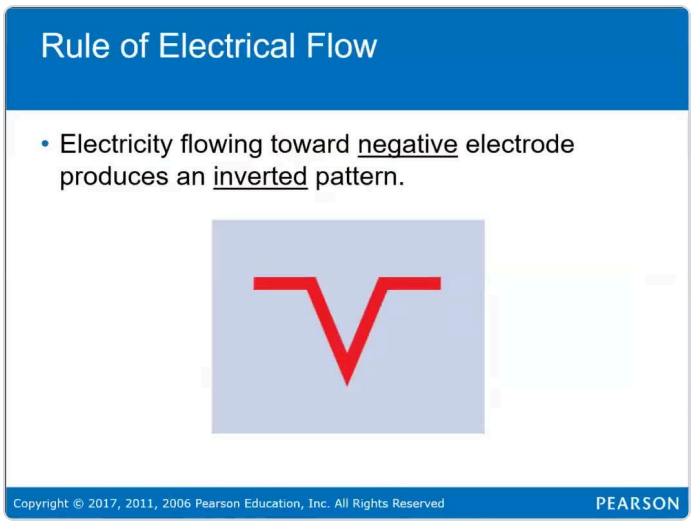
1 · Electrical flow & Lead II | 2 · EKG graph paper | 3 · Box math | 4 · Leads & 12-lead review
5 · The EKG complex | 6 · Cardiac output | 7 · Measuring PRI & QRS | 8 · Intervals & segments
9 · ST segment & MI | 10 · Regularity | 11 · Refractory periods & pulse | 12 · QRS shapes & artifact
Cheat sheet | Quiz

1 Electrical flow review & monitoring Lead II

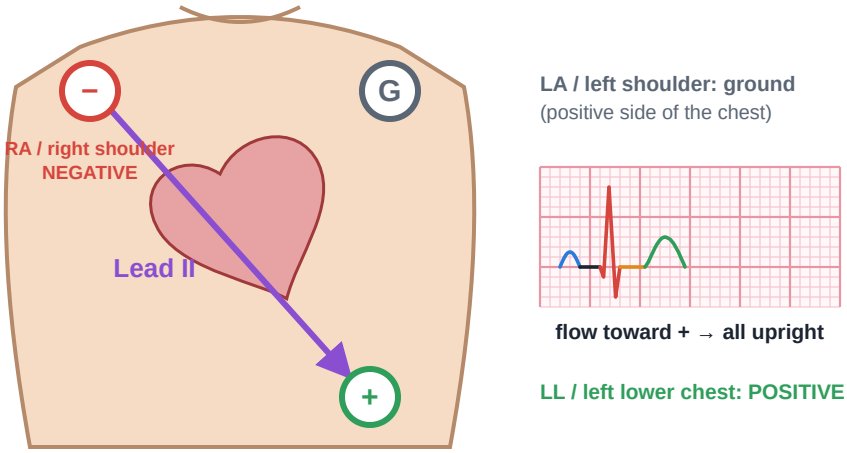
Big idea: current flowing **toward a positive** electrode draws an **upright** wave; toward a **negative** electrode draws an **inverted** (upside-down) wave. The flat middle line is the **isoelectric line** (neutral).



Slide: toward + = upright; toward − = inverted.



Slide: a downward P wave is called a negatively deflected or **inverted** P wave.



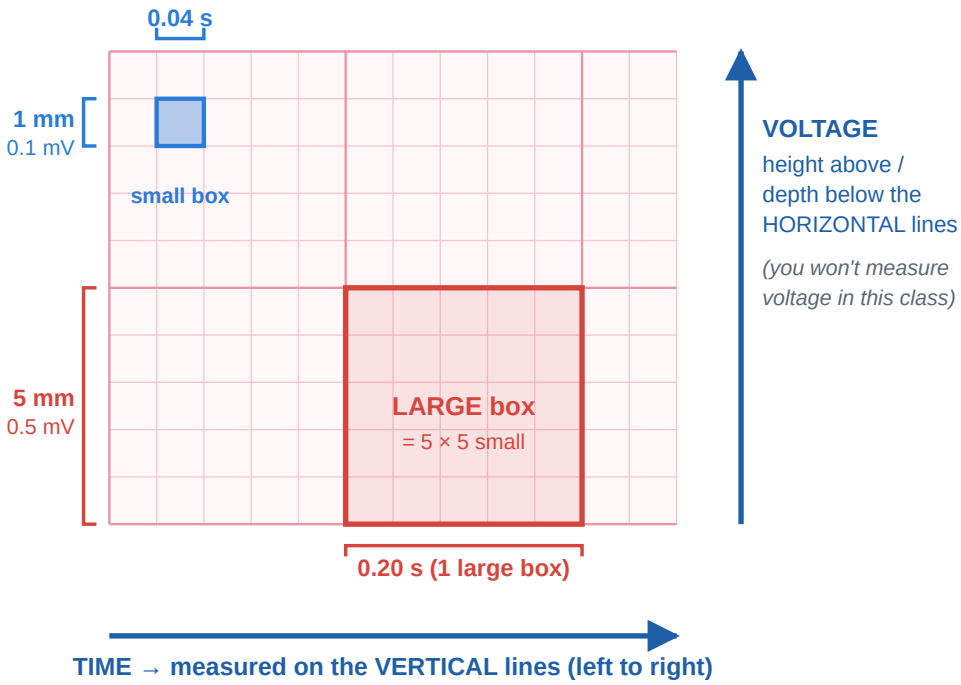
Monitoring Lead II: right shoulder is negative, left leg is positive, left shoulder is the ground. Current runs from − toward +, so P, QRS and T are all upright.

WHAT THE INSTRUCTOR STRESSED (0:00:22–0:02:17)

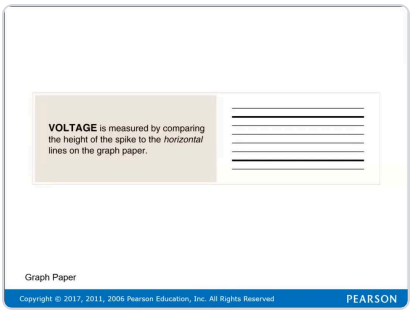
The **right arm is the negative electrode**. Anything flowing toward the left arm or left leg gives a positive (upward) deflection. Above the isoelectric line = positive, below it = negative.

2 EKG graph paper: time & voltage

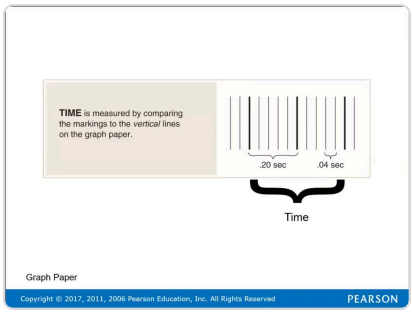
Big idea: left to right is **TIME** (vertical lines); up and down is **VOLTAGE** (horizontal lines). You'll measure **time** by counting boxes all semester.



Zoomed graph paper. 1 small box = 0.04 s; 1 large box (bold pink lines) = 5 small = 0.20 s.



Slide: VOLTAGE is the height compared to the horizontal lines.



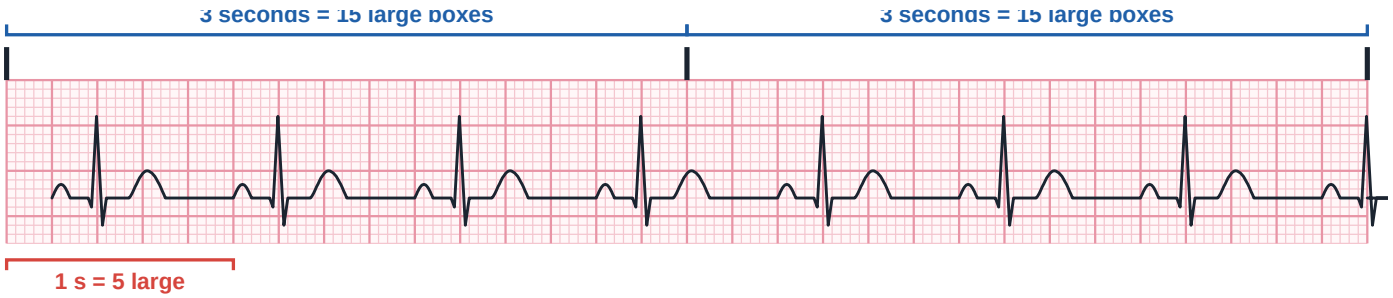
Slide: TIME is read from the vertical lines (0.20 s large, 0.04 s small).

VOLTAGE, IN PLAIN WORDS

The height of the P wave shows how much voltage it took for the atria to depolarize. The height of the QRS (Q down, R up, S down) shows the ventricles' voltage. You **won't** be asked to measure voltage in this class (0:06:27).

3 Box math: the six-second strip

Big idea: the hash marks along the top are **3 seconds apart**. Three hash marks = a **6-second strip**.



hash mark → hash mark → hash mark = 6-second strip = 30 large boxes = 150 small boxes

A 6-second strip. Hash marks are 3 s apart (15 large boxes each). Five large boxes = 1 second.

This many…	…equals
1 small box	**0.04 s**
5 small boxes = 1 large box	**0.20 s**
5 large boxes (25 small)	**1 second**
15 large boxes (hash → hash)	**3 seconds**
30 large boxes (150 small)	**6 seconds**
300 large boxes (1,500 small)	**1 minute**

CAREFUL: 25, NOT 15

In class the count was first said as "15 small boxes to a second" and then corrected: **5 large × 5 small = 25 small boxes = 1 second**.

LIKELY EXAM QUESTION (0:05:00)

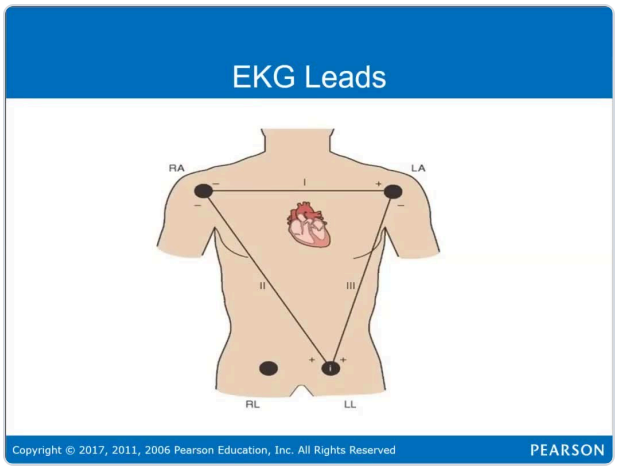
"Maybe I need to add… **how many boxes make up a minute?**" Work it out: 5 large/s × 60 s = **300 large**; 25 small/s × 60 = **1,500 small**. A calculator is allowed on the test.

MEMORY TRICK

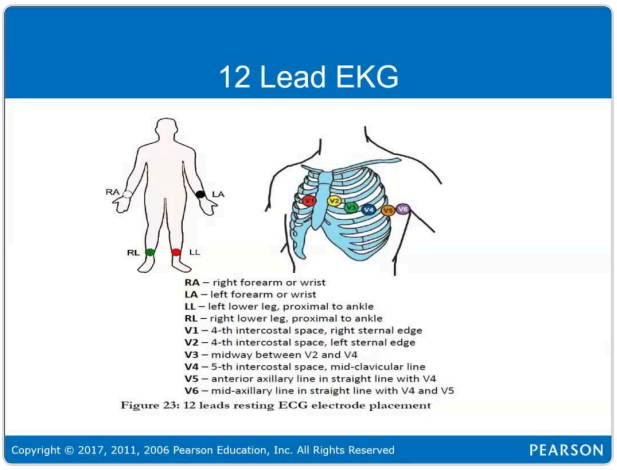
"**Five-five-one**": **5** small make a large, **5** large make **1** second. Count up by 0.2: 0.2, 0.4, 0.6, 0.8, **1.0**.

4 Leads & 12-lead review (from Week 2)

Big idea: a **lead** = two electrodes, usually on opposite sides of the heart; the EKG records the flow of electricity between them. **10 electrodes → 12 views** of the heart.

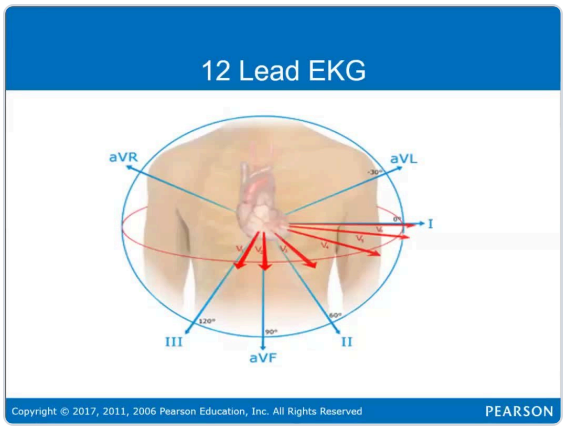


Einthoven's triangle: I = RA→LA, II = RA→LL, III = LA→LL. RA is negative; LA and LL are positive.



12-lead placement: limb electrodes on the wrists and ankles, V1–V6 on the chest.

Electrode	Where it goes (patient's own left/right)
V1	4th intercostal space, **right** of the sternum
V2	4th intercostal space, **left** of the sternum
V4 ***(place before V3)***	5th intercostal space, left **midclavicular** line
V3	Halfway between V2 and V4
V5	5th intercostal space, **anterior axillary** line (front of the armpit)
V6	5th intercostal space, **mid-axillary** line (middle of the armpit)



Lead axes: blue = frontal plane (limb + augmented leads); red = horizontal plane (V1–V6).

CLINICAL: KEEP PLACEMENT CONSISTENT

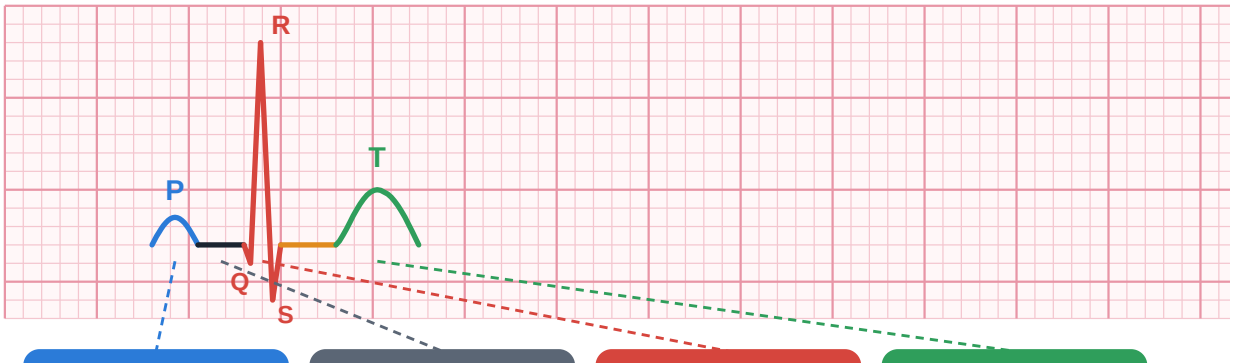
Patients often get a 12-lead every day. Comparing today's EKG with yesterday's works only if the electrodes go in the same spots. If a sternotomy dressing or an amputation is in the way, move the electrode, and keep using that spot.

TRANSCRIPT SLIP

At 0:11:59 V4 is said to go at the "4th intercostal space at the right sternal edge". The slide and Week 2 both say **5th ICS, left midclavicular line**. Use the slide.

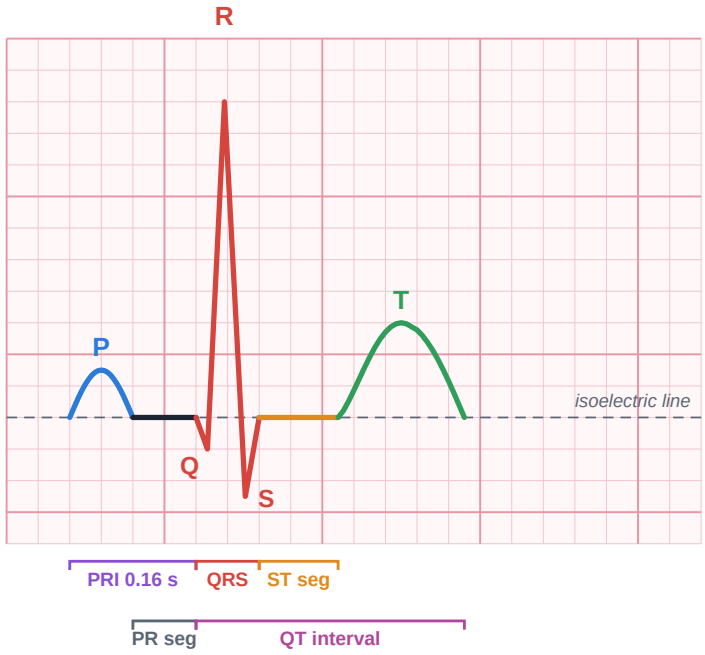
5 The EKG complex: one heartbeat on paper

Big idea: every wave is a step of the conduction pathway: P atria depolarize → pause at the AV node → QRS ventricles depolarize → T ventricles repolarize.



P wave	PR segment	QRS complex	T wave
SA node fires (primary pacemaker) → interatrial pathway → atrial muscle **ATRIA DEPOLARIZE**	internodal pathways → AV node **HOLDS the impulse** (short delay so atria fill the ventricles)	Bundle of His → R & L bundle branches → Purkinje fibres → ventricles **VENTRICLES DEPOL.**	ventricles **REPOLARIZE** (recover). Atrial repol. is hidden in the QRS

Conduction pathway lined up with the waveform it makes.



P wave — atrial depolarization

PR segment — AV-node delay

PR interval — P start → QRS start

QRS complex — ventricular depol.

ST segment — QRS end → T start

T wave — ventricular repol.

QT interval — QRS start → T end

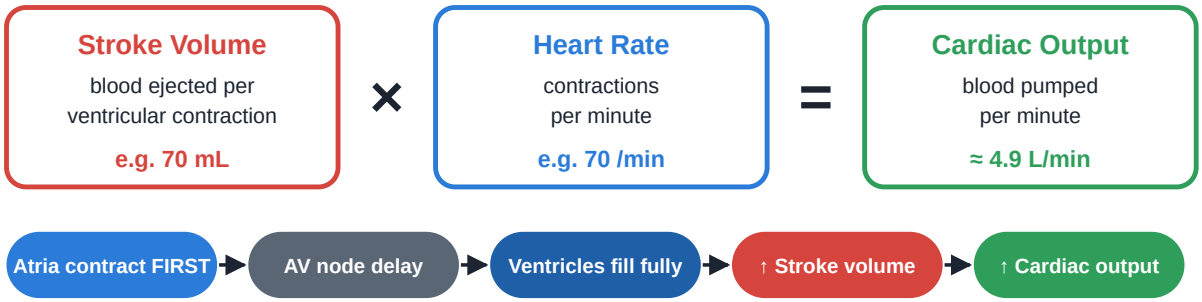
One complete heartbeat with every named part. Measurements here: PRI 4 small boxes = 0.16 s; QRS 2 small boxes = 0.08 s.

WORDS TO KNOW

In class, "depolarization" was used as another word for contraction. Strictly, depolarization is the **electrical** event that **triggers** the muscle to contract. **Atrial repolarization never shows**: it's hidden in the QRS.

6 Why top-down matters: cardiac output

Big idea: **Cardiac output = Stroke volume × Heart rate.** The AV-node delay lets the atria top up the ventricles first, which increases stroke volume.

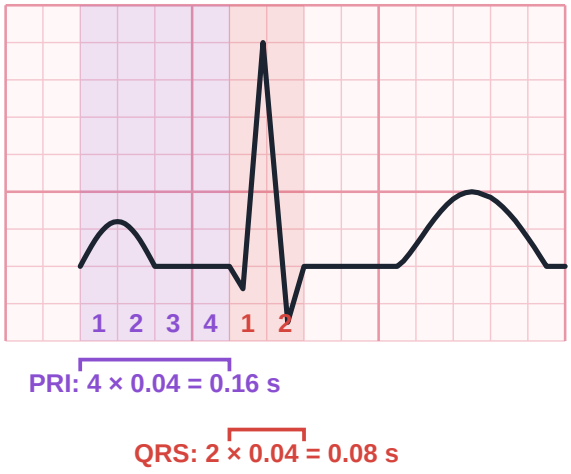


You should know this formula (0:15:50). Example: 70 mL × 70 beats/min = 4,900 mL/min ≈ 4.9 L/min.

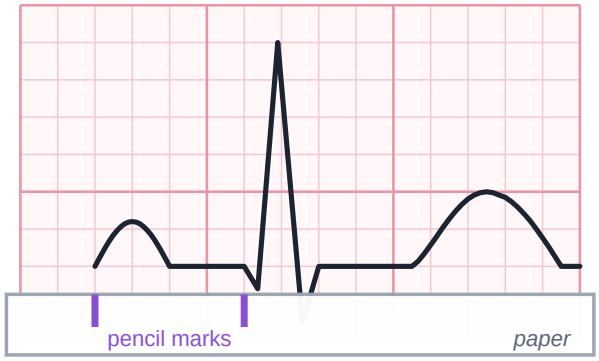
7 Measuring the PRI and QRS: count small boxes

Big idea: find where the wave **starts** and **ends**, count the small boxes between them, and multiply by **0.04 s**.

① Count the small boxes



② Trick: mark it on scrap paper

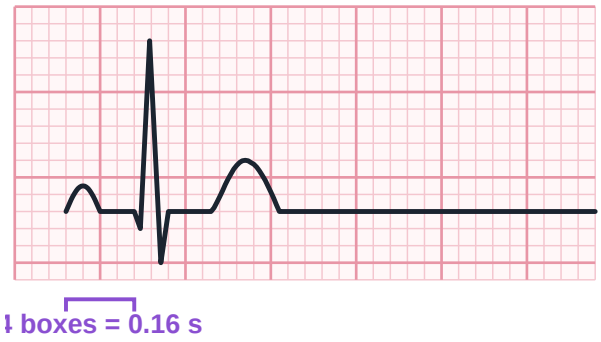


③ Slide it onto clean grid and count:

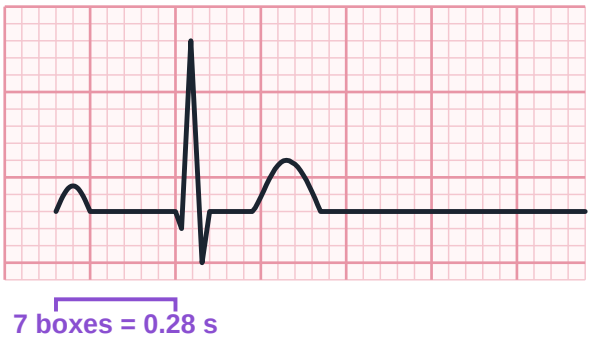


Instructor's worked example (0:17:52–0:20:14): PRI = 4 small boxes = 0.16 s; QRS = 2 small boxes = 0.08 s. Mark both ends on scrap paper, then count on a clean part of the grid.

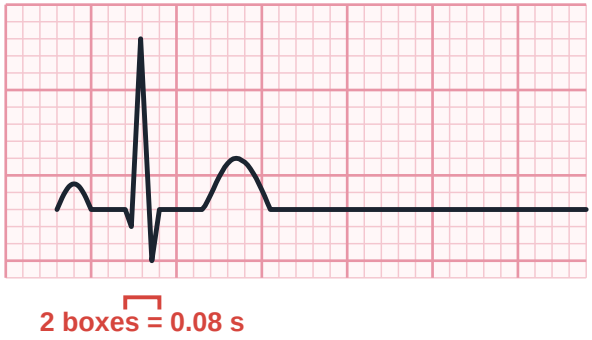
PR interval — normal ✓ 3–5 boxes



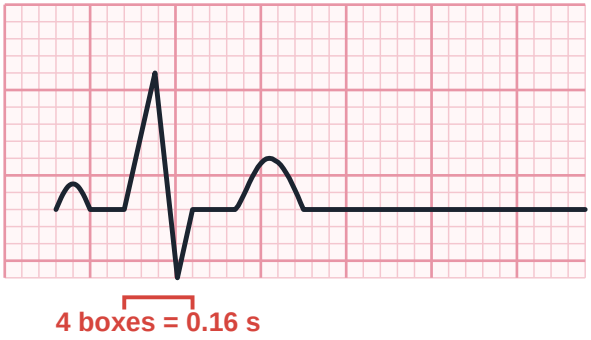
PR interval — too long ✗ > 5 boxes



QRS — narrow (normal) ✓ ≤ 3 boxes



QRS — wide ✗ > 3 boxes



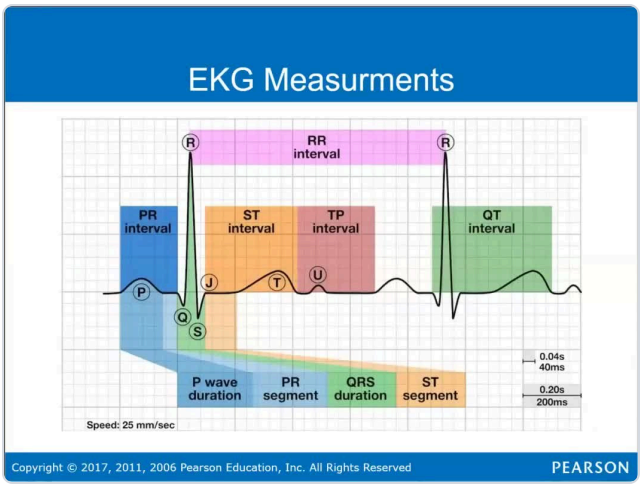
WHAT YOU WILL AND WON'T MEASURE (0:17:52, 0:21:36)

You **will** measure the **PRI** and the **QRS width**. You won't measure the PR segment, ST segment or QT interval; a 12-lead machine measures those automatically and compares them to normal values.

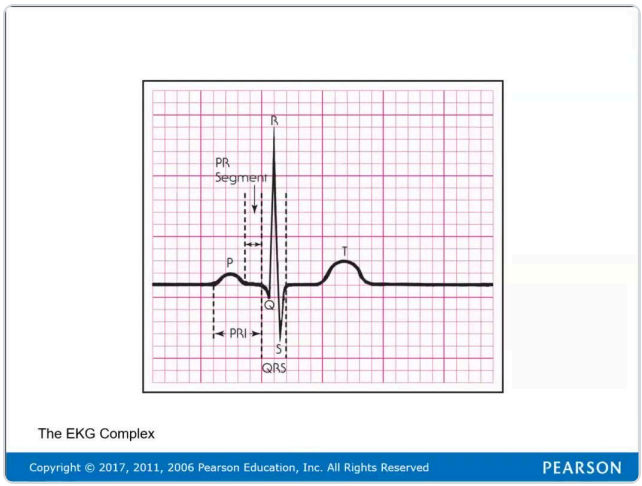
8 Intervals & segments: definitions

> **Big idea:** an **interval** includes at least one wave; a **segment** is the flat line **between** waves.

Measurement	From → to	What it shows	Normal
PR interval	start of P → start of QRS	atrial depolarization + AV-node delay	**0.12–0.20 s** (3–5 small boxes)
PR segment	end of P → start of QRS	delay at the AV node	not measured
QRS complex	start of Q → end of S	ventricular depolarization	**≤ 0.12 s** (≤ 3 small boxes)
ST segment	end of S (J point) → start of T	between ventricular depol. & repol.	flat, at the isoelectric line
QT interval	start of QRS → end of T	all of ventricular systole (depol. + repol.)	machine measures it



Slide: colour map of each interval and segment (RR, PR, ST, TP, QT).



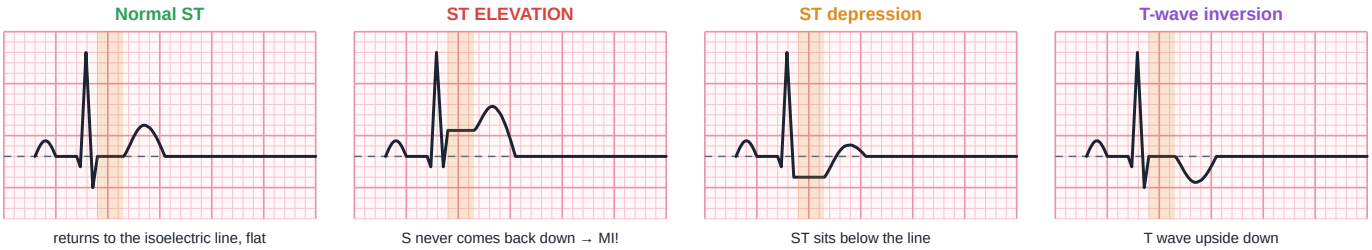
Slide: The EKG complex with the PRI and PR segment marked.

SMALL WORDING DIFFERENCE

At 0:21:36 and 0:22:50 the ST segment is described as going from the S wave "to the end of the T wave". The slide graphic calls that span the **ST interval**. The **ST segment** is the flat part from the end of the QRS to the **start** of the T wave. Either way, what matters is **whether it comes back to the isoelectric line**.

9 The ST segment & a heart attack (MI)

Big idea: a normal ST segment returns to the isoelectric line. In an active **myocardial infarction** the S wave stops short, runs straight across and climbs into a tall T wave: this is **ST elevation**.



Orange band = ST segment. ST elevation is the most common 12-lead finding in an active MI.

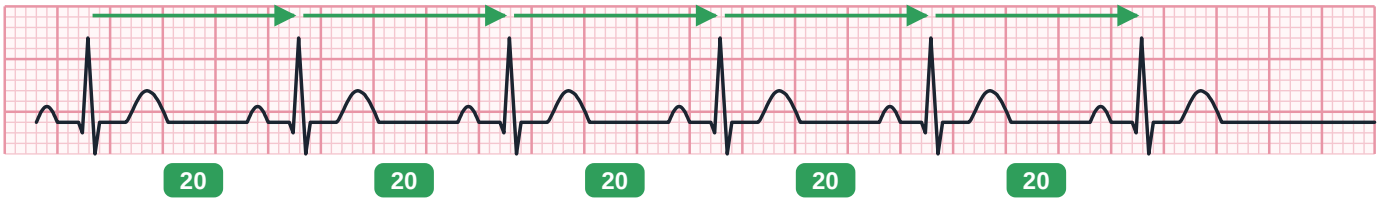
CLINICAL SCENARIO (0:24:21)

A patient suddenly has a heavy chest, jaw, back or radiating arm pain, shortness of breath or faintness. Put on oxygen, **get a 12-lead EKG** and draw blood (troponin: you don't need to know that for this test).

10 Is the rhythm regular? Count R to R

Big idea: count the small boxes between consecutive **R waves** all the way across the strip. Same count each time = **regular**; changing count = **irregular**.

REGULAR — same number of small boxes between every R wave



IRREGULAR — spacing changes across the strip



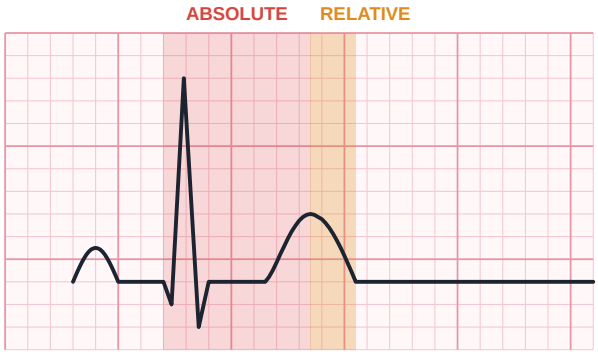
Numbers = small boxes between R waves. Top: 20-20-20-20-20 (each R-R = 0.80 s). Bottom: 14-25-17-28-18.

COMING NEXT (0:25:23)

Regularity is the first step of the **systematic approach** to every rhythm strip (Chapter 3). Get fast at counting boxes now.

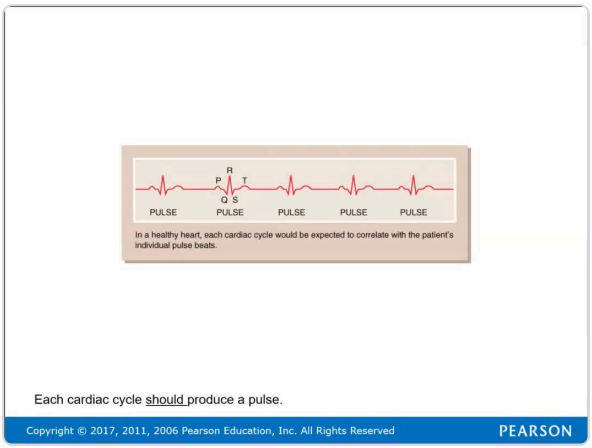
11 Refractory periods & "each cycle should make a pulse"

Big idea: right after depolarizing, heart cells need time to recover. They are **absolutely** unresponsive first, then **relatively** responsive.



Refractory periods (only named in class, 0:25:49).

- **Absolute**: cells can't respond to ANY stimulus (QRS start → about the peak of T).
- **Relative**: a STRONG stimulus can fire them early (downslope of T), the "vulnerable" spot.



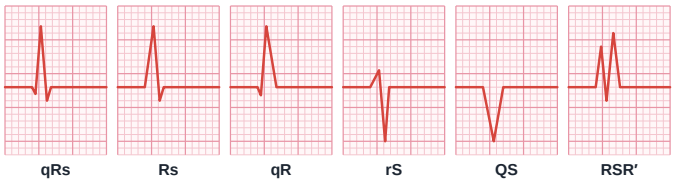
Slide: in a healthy heart every complex matches a pulse beat.

CHECK THE PATIENT, NOT JUST THE MONITOR

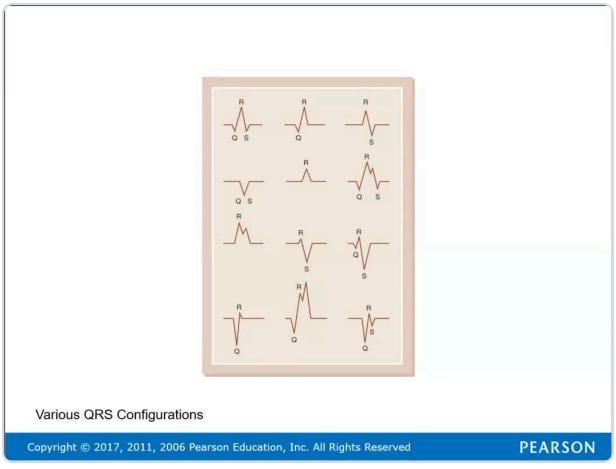
The EKG shows **electrical** activity only. Feel for a pulse to confirm that each complex really pumped blood.

12 QRS shapes & artifact

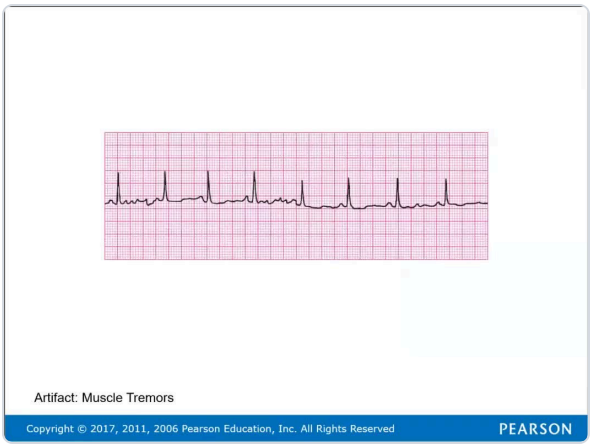
Big idea: not every QRS has all three waves, but it's still called a "QRS complex". **Artifact** is junk on the tracing that doesn't come from the heart.



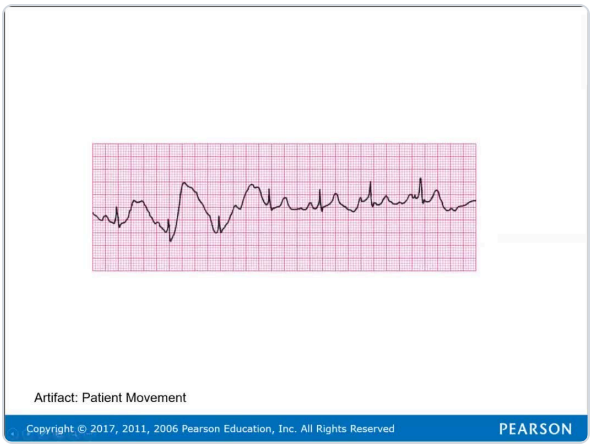
Naming: Q = first DOWNWARD wave before R; R = first UPWARD wave; S = downward wave after R; R′ = a second upward wave. Big waves get CAPITALS, small ones lowercase.



Slide: various QRS configurations (shown without commentary).



Artifact from MUSCLE TREMORS: fine, jittery baseline.



Artifact from PATIENT MOVEMENT: big wandering swings.

SPOTTING ARTIFACT

Look for the real QRS complexes marching through the noise at a steady rate. Fix it by warming and calming the patient, checking electrodes and asking them to stay still. (These three slides were shown without commentary; textbook Ch. 2.)

★ Cheat sheet: Week 3 on one page

Paper

Unit	Value
Small box	0.04 s · 1 mm · 0.1 mV
Large box (5 small)	0.20 s · 5 mm · 0.5 mV
1 second	5 large = 25 small
Hash mark → hash mark	3 s (15 large)
3 hash marks	6-second strip
1 minute	300 large = 1,500 small
Horizontal lines	voltage (height)
Vertical lines	time (width)

Electrical flow

Toward + electrode	upright
Toward − electrode (RA)	inverted
Right angle to lead	isoelectric
Lead II	RA (−) → LL (+), all upright

Waves

Wave	Meaning
P	atrial depolarization
PR segment	AV-node delay
QRS	ventricular depolarization
ST	pause; elevated in MI
T	ventricular repolarization
(atrial repol.)	hidden in the QRS

Normals to memorise

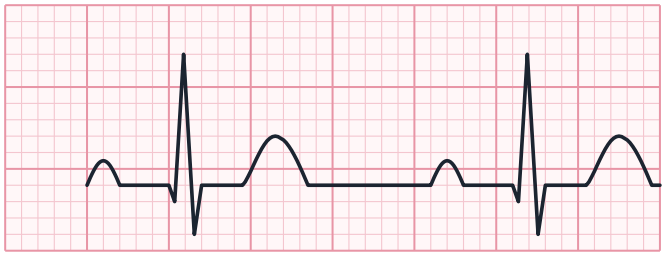
PRI	0.12–0.20 s = 3–5 small boxes
QRS	≤ 0.12 s = ≤ 3 small boxes
Regularity	R-R counts all equal?
CO	SV × HR

Conduction order

SA node → interatrial & internodal pathways → **AV node (delay)** → Bundle of His → R & L bundle branches → Purkinje fibres → ventricular muscle

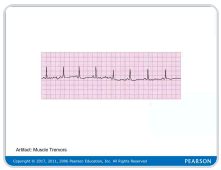
? Self-test quiz

1. One small box equals ___ s and one large box equals ___ s.
2. How many **small** boxes make one second? How many **large** boxes make one **minute**?
3. Three hash marks along the top of a strip mark off how many seconds?
4. Do the horizontal lines on the paper measure time or voltage?
5. Current is flowing toward the right-arm electrode. Which way does the wave deflect, and why?
6. Name what each represents: P wave, PR segment, QRS complex, T wave.
7. Why does the AV node hold the impulse briefly? Link your answer to cardiac output.
8. Write the cardiac output formula. If SV = 60 mL and HR = 80/min, what is the CO?
9. **Measure this strip:** the P wave starts on a dark line. Find the PRI and the QRS width (in boxes and seconds). Are they normal?



Quiz strip (1 small box = 0.04 s).

10. The R-R spacing on a strip is 20, 20, 20, 20 small boxes. How many seconds is each R-R, and is the rhythm regular?
11. Where does V4 go, and why do you place V4 before V3?
12. Which finding on a 12-lead is the most common sign of an active MI? Describe what it looks like.
13. **Identify:** what is causing the fuzzy, jittery baseline on this strip?



14. Which two measurements will you be asked to measure by hand this semester?

Answers. 1) 0.04 s; 0.20 s. 2) 25 small per second; 300 large per minute (= 1,500 small). 3) 6 seconds (marks are 3 s apart). 4) Voltage (height); vertical lines = time. 5) Downward (inverted), because the right arm is the negative electrode. 6) P = atrial depolarization; PR segment = AV-node delay; QRS = ventricular depolarization; T = ventricular repolarization. 7) So the atria contract first and top up the ventricles; fuller ventricles mean a larger stroke volume, so a larger cardiac output. 8) CO = SV × HR = 60 × 80 = 4,800 mL/min ≈ 4.8 L/min. 9) PRI = 5 small boxes = 0.20 s (upper limit of normal); QRS = 2 small boxes = 0.08 s (normal). 10) 20 × 0.04 = 0.80 s each; regular. 11) 5th ICS, left midclavicular line; V3 goes halfway between V2 and V4, so you need V4 in place first. 12) ST elevation: the S wave doesn't come back to the isoelectric line, the ST runs straight across into a high T wave. 13) Muscle-tremor artifact. 14) PR interval and QRS width.